

Fall 2026

**CISC/CMPE 320 : Fundamentals of
Software Development**

1

Topic: Setting up a C++ Environment

Note Taking

- noteQ is a new note taking tool that has been designed and created by Queen's Student Accessibility Services in collaboration with Queen's IT Services
- The noteQ process integrates with Ventus and onQ to allow students registered with QSAS, and who have been provided with note taking to access class notes shared by their fellow classmates.

Note Taking

- A message has been posted to the CISC 320 001 – Fall 2026 course announcements
- If, after 2 weeks, no notes have been uploaded to noteQ, a second message will be posted on the CISC 320 001 - F26 course announcements.
- Other announcements may be added during the course to remind/request students to share their notes.

Note Taking

- If you have further questions about noteQ, please contact notetake@queensu.ca. If a student does approach you to you about how to ask about Note Taking, please redirect them to notetake@queensu.ca.

Course Project

1. Have you been assigned to a group ?
2. If Yes:
 3. Start thinking about a project
 4. Discuss your project in 2nd week (team)
 5. Exit
6. If No:
 7. Please do;
 8. Then do 2 above.

C++

- General-purpose programming language.
- An extension of the C programming language.
- Has object-oriented and generic programming features.
- Is multi-platform (Windows, Linux, Mac)

What do you need to run a simple C++ program?

- A text editor:
 - Any will do for simple programs.
 - Examples of **basic editors**: Sublime, VS Code
 - Examples of **sophisticated editors** that offer extra features (i.e. IDEs):
Eclipse, Code blocks, **Xcode**, CLion.
- A C++ compiler
 - a computer program which converts a high-level language into a machine understandable low-level language (assembly language).
 - Platform dependent.
 - **GCC**

What's GCC?

- Stands for GNU Compiler Collection
 - GNU is a recursive acronym for "GNU's Not Unix!"
 - GNU is a large collection of free software
- GCC (a subset of GNU) is an integrated distribution of compilers for several major programming languages, one of which is C++.
- `g++` is the invocation command used for preprocessing, compilation, assembly and linking of source code to generate an executable file
 - `g++ fileName.cpp -o fileName.out`
 - `./filename.out`

More on compilers

- If you are using a **Windows** machine:
 - Opt for MinGW as the compiler.
 - MinGW is a compiler system based on the GNU, GCC that compiles and links code to be run on Win32 (Windows) systems.
- If you are using a **Mac**:
 - You can install GCC by installing Command Line Tools for Xcode using the command:
 - `xcode-select --install`

More on IDEs

- CrossPlatform(recommended):
 - Visual Studio Code (VS Code)
 - JetBrains CLion (free for students)
 - Eclipse for C++
- If you are using a **Windows** machine:
 - Code::Blocks is ok.
- If you are using a Mac:
 - Xcode is a good **choice**.

Let us try it.

- *HelloWorld* demo running on a Mac terminal using the **Sublime** editor and the **g++** compiler.
- Here is the code:

```
#include <iostream>

int main(int argc, const char * argv[]) {
    // insert code here...
    std::cout << "Hello, World!\n";
    return 0;
}
```

Let us try it.

- Open text editor
- Write the code
- Save with .cpp extension
- Open terminal
- Navigate to directory where the file is stored
- Run the following commands:
 - `g++ HelloWorld.cpp -o HelloWorld.out`
 - `./HelloWorld.out`

For more complex code

- A proper IDE is needed.
- A recommended safe configuration that causes less trouble during the marking process is usually **Eclipse + MinGW**.
- A few steps are needed to link the two, i.e. to tell Eclipse where it can find the MinGW compiler to run the C++ program.

Installing MinGW

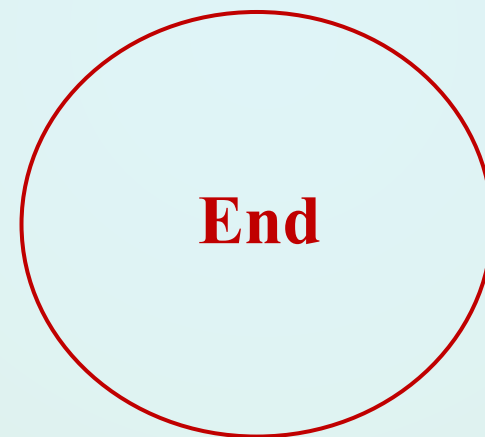
1. Download MinGW.
2. Extract archive into C:\MinGW (actually to C:\, because archive contains folder MinGW)
3. Go to This PC -> Properties -> Additional system settings -> **Tab** Advanced-> **Button** Environment variables
4. On the second table System variables click New. Name the variable MINGW_HOME and set path to the MinGW install folder C:\MinGW and then OK
5. Find the variable Path in the table and choose Edit
6. In the new window click New and type %MINGW_HOME%\bin\
7. Confirm actions by clicking OK in the open windows
8. You can check that it worked by typing in the command line g++ -version.

Installing Eclipse

1. Install Eclipse with CDT or just add CDT to existing Eclipse installation
2. Go to the folder with installed MinGW (C:\MinGW\bin\). Make a copy of the file gcc.exe (DO NOT RENAME original file!)
3. Rename copied file to mingw32-gcc.exe (You should have both files gcc.exe and mingw32-gcc.exe in \MinGW\bin\)
4. Open Eclipse and select the C\C++ perspective
5. Go to Window -> Preferences -> C\C++ -> Build -> Environment
6. Click Add and type PATH as the name and click on Variables and select Path. Confirm with Ok.
7. Select new variable PATH by clicking Select and then Ok.
8. Restart Eclipse

Make sure your environment is ready very soon!

- In OnQ forum you can discuss common issues you may face during installation.
- Please use it and share your experience. In case you faced an issue, please explain to your classmates how you solve it.
- Windows Vs. Apple
Please start your discussion by mentioning one at the first line solely.



Next: History

